SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks.* *(BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I HARINI G/192524011_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Harinigopi

Annexure A

CONCEPT MAPPING

1.A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

ANSWER:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Explanation

Session Layer

- Establishes, manages, and terminates communication sessions between devices.
- Keeps the connection active during data transfer.
- Synchronizes communication and allows recovery if interrupted.

Data Link Layer

- Converts packets into frames.
- Uses **MAC addresses** for local device-to-device communication.
- Detects transmission errors using **CRC**.
- Controls access to the physical transmission medium.

Importance for Reliable Communication

- The **Session Layer** ensures that communication between applications remains organized and uninterrupted.
- The **Data Link Layer** ensures data is transmitted accurately over the local network by detecting errors and delivering frames to the correct device.
- Together with the other OSI layers, they provide reliable, efficient, and error-free communication across networks.

2.Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

ANSWER:

Application Layer → Transport Layer → Segment→ Network Layer→ Packet → Data Link Layer → Frame → Physical Layer → Bits

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

ANSWER:

Data Link Layer
|
MAC Address
|
Local Delivery

Network Layer
|
IP Address
|
Global Delivery

Explanation

- Data Link Layer uses the MAC address for local delivery within the same network.
- Network Layer uses the IP address for global delivery between different networks.
- Together, the IP address routes the data to the correct network, and the MAC address delivers it to the correct device, ensuring complete data delivery.

4.A concept map shows:

Error Control

→Data Link Layer → Error Detection→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

ANSWER:

- The Data Link Layer detects errors in frames during transmission using CRC (Cyclic Redundancy Check).
- It ensures reliable communication between two directly connected devices.
- The Transport Layer provides end-to-end communication between the sender and receiver.
- It uses acknowledgements (ACKs) and retransmission to recover lost or corrupted data.
- The Data Link Layer handles local error detection, while the Transport Layer handles end-to-end error recovery.
- Together, these layers ensure that data is delivered accurately, completely, and reliably with minimal errors.

5.Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

ANSWER:

User Opens Browser →Application Layer (HTTP/HTTPS) → Presentation Layer (Encryption) → Session Layer (Session) → Transport Layer (TCP) →Network Layer (IP) →Data Link Layer (Frames) → Physical Layer (Signals) →Website Server.

6.A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✕ → Transport ✕ → Application ✕

Analyze the problem and explain how the concept map helps in troubleshooting.

ANSWER:

- The Physical Layer ✓ indicates that the cables, Wi-Fi, or other hardware connections are functioning properly.
- The Data Link Layer ✓ confirms that data frames are being transmitted correctly within the local network.
- The Network Layer ✕ indicates a problem with IP addressing, routing, or router configuration.
- Since the Network Layer has failed, the Transport Layer ✕ cannot establish end-to-end communication, and the Application Layer ✕ cannot access network services such as web browsing or email.
- This concept map helps in troubleshooting by identifying the exact layer where the failure occurs. Instead of checking every component, the network administrator can focus on the Network Layer, making the troubleshooting process faster and more efficient.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation